

Topic 5 - Arms Race

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Introduction

This section should outline the real-world background and context of your model.

Discuss the motivation one might have to study this model. You may want to research the relevant literature and include some citations to support your framing of the real-world context (depending on the project). Be sure to include only background information and context that is directly related to your research project. Really focus on what the reader needs to know to understand your work, and be interested and excited by it.

The final paragraph of your Introduction should frame your research question and describe what you will be studying in your paper. Briefly mention, in words, what you will be doing and showing in the rest of the paper. It is often good to lay out your main finding/result here, too. A good Introduction section for an applied mathematics paper ideally has no mathematical symbols or jargon; instead, it should be focused on the real-world context and what you are trying to study. The math is a tool you will introduce later, not the main focus of the paper.

Base Model

Start by explaining the real-world mechanisms you're modelling, and any you're omitting. Discuss the assumptions you're making in reducing the real-world system to a mathematical model. Define the state variables you'll be working with. Don't forget to clearly state the units of the variables.

Define parameters (and their units) and cite any sources for the values you're using to parameterize your model. If you cannot find specific parameter values in the literature, make educated guesses for values and explain why you're using these particular values (in this case, you may want to consider a range of parameters instead of just one specific value to ensure your modelling results are robust to some uncertainty in parameter values).

You may want to include a model diagram (e.g., if it's a compartmental model). Write out the model equation(s) and explain any modelling choices you've made in them (e.g., discrete- vs. continuous-time)

Analysis of Base Model

This subsection should describe the various analysis you've performed on your base model and the results you've obtained from them. You should be using a mix of analytical tools (e.g., finding equilibrium points, determining stability) and numerical tools (e.g., model simulations in R). When discussing analytical calculations, you don't need to list out every single step of your calculation; just present the important/key steps. When discussing numerical simulations, you should describe what your R code is doing and present some plots to supplement

your written descriptions; all R code should be presented in an appendix at the end of the paper (see below). Be sure to describe what these analysis tell you about your base model and what predictions they make (this should be in the context of the real-world scenario framing your project).

Model Extension

Here, you should describe the extension you've chosen to study for your model. Start by discussing what the extension is and how you plan to change your base model to account for it. Similar to the base model subsection, be sure to talk about any assumptions in your modelling, define state variables and parameters (including units), and describe any modelling choices made. In particular, you should highlight how your extended model differs from your base model. If appropriate, it would be good to include a model diagram. You should also write out the model equations. Do not discuss any analysis of the extended model here. That will go in the next section

Results

In this section, you should describe the analysis you performed on your extended model. Be sure to motivate your analysis in the context of your research question (i.e., why do you want to perform each analysis, what are you hoping it will tell you). These analysis can be purely simulation-based: you are not required to perform any mathematical analysis of your extended models, but if you can, you should! Mathematical analysis generally tell you more about model behaviour than

simulations (when both are possible). You should be including plots here, as well, to go along with any simulations you did. Unlike with the draft, this section of the paper should now be complete and all results should be discussed in the paper.

Discussion

This is the final section of the main body of your paper. Start by briefly summarizing the main results you presented in the last section and relating them back to your research question. This is also where you should be providing answers, based on your results, to your research question(s). Be sure to interpret your results in the context of the real-world scenario framing your project. What are the implications of your results?

You should also discuss limitations of your work. This is usually related to the modelling assumptions you made before. When are your results applicable and when are they not applicable? What factors have you omitted from your model that may affect the predictions? This is important to discuss, otherwise the reader may come away from your paper with the impression that your results are universal which is rarely the case.

Finally, you should discuss possible directions for future research. Did your results open up any other questions? Where else could you go with this work? Are there other things you've read about in the literature that could be interesting to look at? You should also use this as a way to conclude your whole paper.

References

Appendix: Individual Contributions